ECON2010H INTERMEDIATE MACROECONOMIC THEORY I

WEEK 02: BUSINESS CYCLE MEASUREMENT

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READING: TEXTBOOK CHAPTER 3

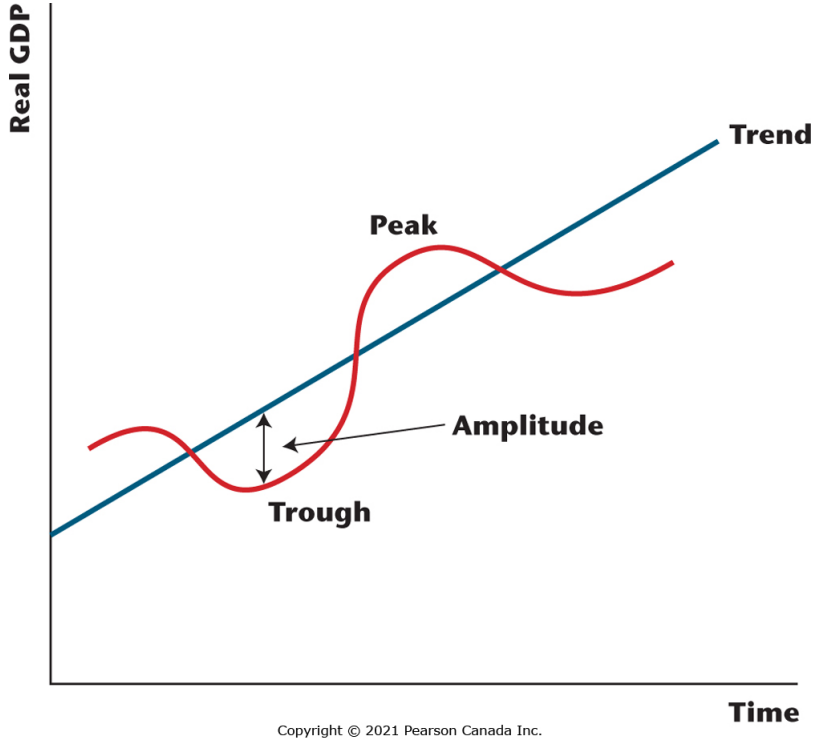
OUTLINE

1. Regularities in GDP fluctuations
2. Co-movements
3. The Phillips curve

1. REGULARITIES OF FLUCTUATIONS OF REAL GDP

- **Business cycles** are fluctuations of real GDP about its trend.
- A **peak** is relatively a large positive deviation from trend real GDP.
- A **trough** is relatively a large negative deviation from trend real GDP.
- **Booms** are persistent positive deviations from trend real GDP.
- **Recessions** are persistent negative deviations from trend real GDP.

ILLUSTRATION OF BUSINESS CYCLES



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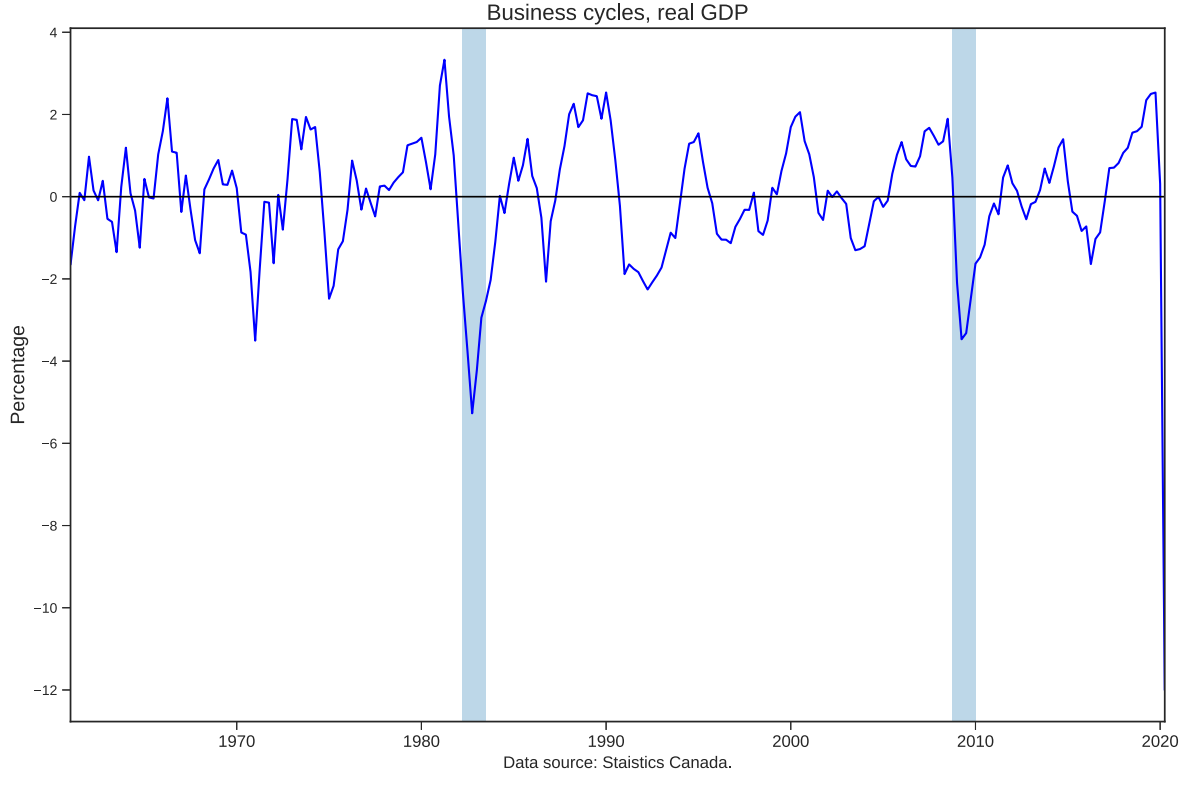
REGULARITIES OF REAL GDP FLUCTUATIONS

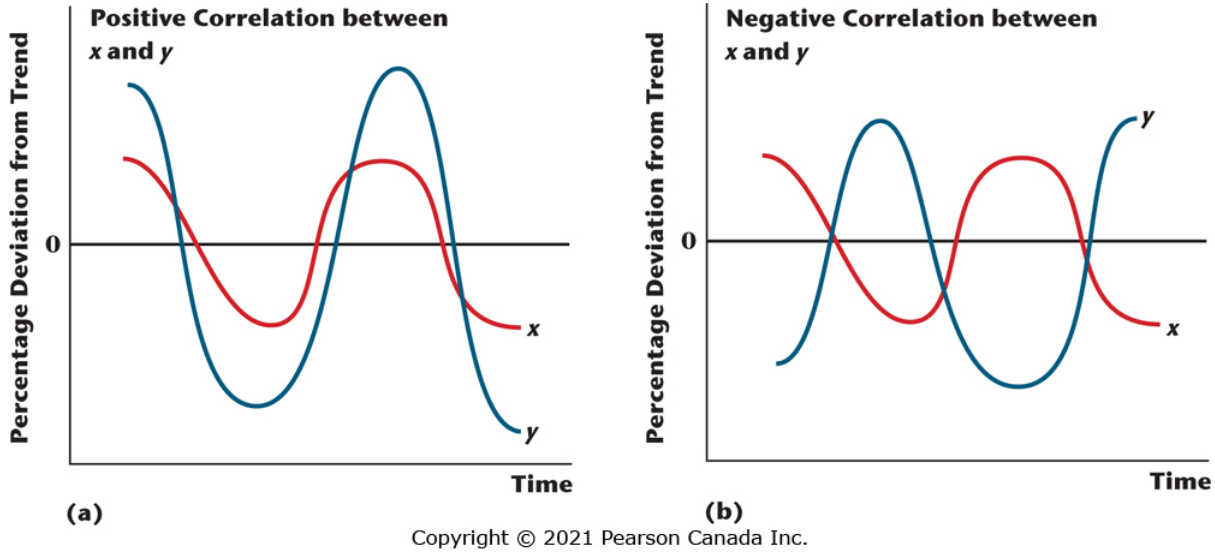
- **Persistence**. When real GDP is above the trend, it tends to stay above trend. When it is below the trend, it tends to stay below trend.
- No regularity in the amplitude (size of deviatoin from trend) of fluctuations.
- The length of time between peaks and troughs varies considerably.
- The persistence of deviation of real GDP from trend makes short-run forecasting relatively easy.
- But, the longer-run forecasting is difficult.
- Predicting future fluctuations in real GDP by looking at the past is like forecasting weather.



- Recall, $\ln Y_t^c = \ln Y_t - \ln \overline{Y}_t$. Figure shows $100 \ln Y_t^c$, deviation from trend in percentage.

2. CO-MOVEMENT

- Co-movement refers to that macroeconomic variables fluctuate together.
- We can use the **correlation coefficient** to measure the degree of co-movement between two variables.
    - Its value is between -1 (perfect negative correlation) and 1 (perfect positive correlation).
    - If the coefficient is positive, fluctutations in two variables qualitatively move together.
    - If the coefficient is negative, fluctuations in two variables qualitatively antedate.
- For your information (FYI), correlation coefficient beween x and y is $\rho_{xy} = \frac{COV(x,y)}{\sigma_x \sigma_y}$.
    - $COV(x, y)$ is the **covariance** between x and y.
    - σ_x and σ_y are respectively the **standard deviation** of x and y.



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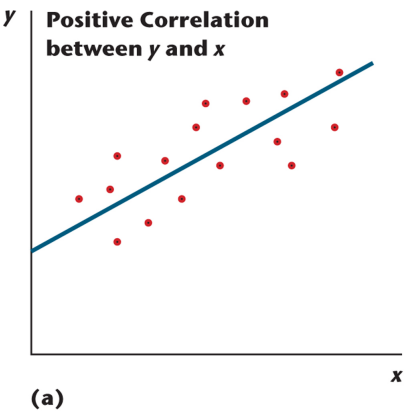
- Another way to detect comovement is the **scatter plot**.



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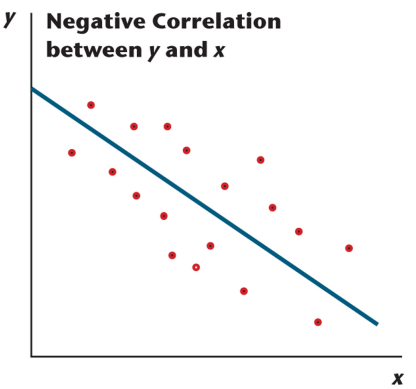

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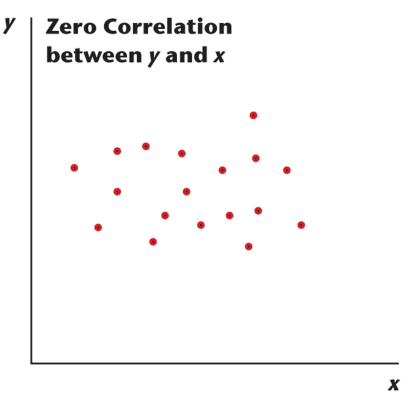

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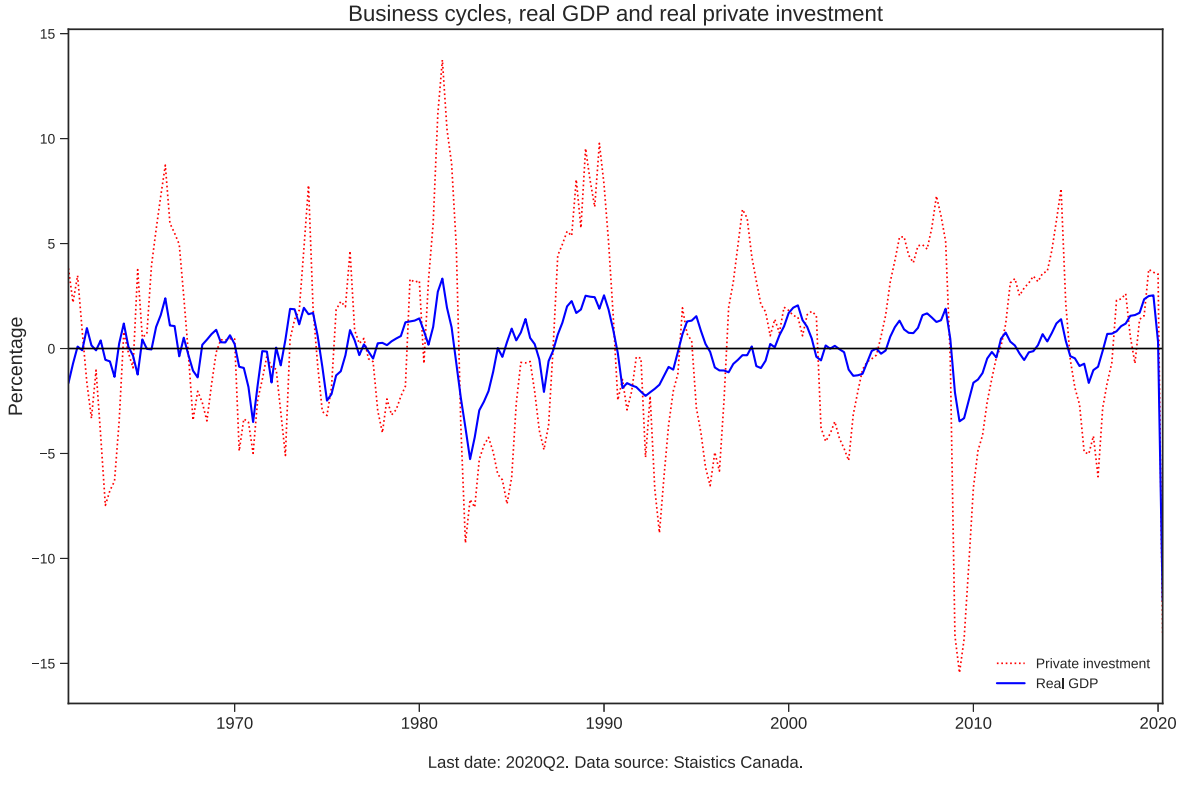
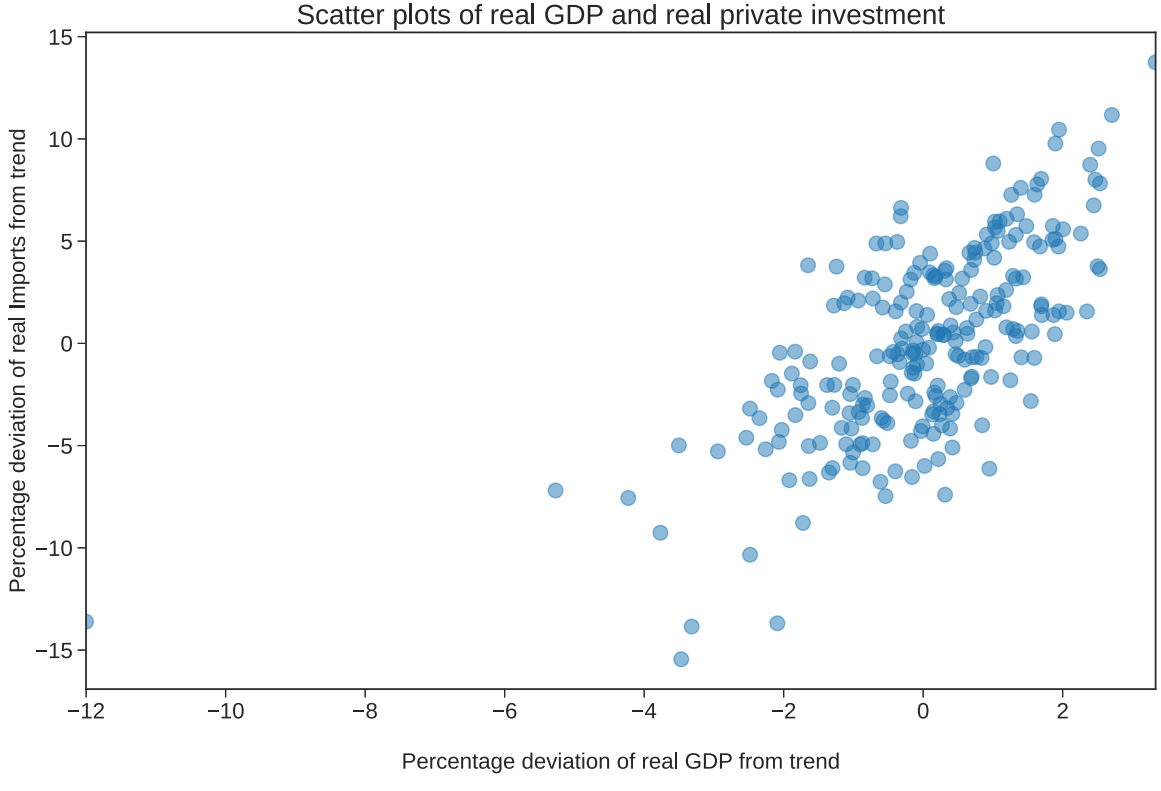
- Example, comovements of real GDP and real private investment




Scatter plots of real GDP and real private investment
Percentage deviation of real Imports from trend
Percentage deviation of real GDP from trend
15
10
5
0
−5
−10
−15
−12
−10
−8
−6
−4
−2
0
2

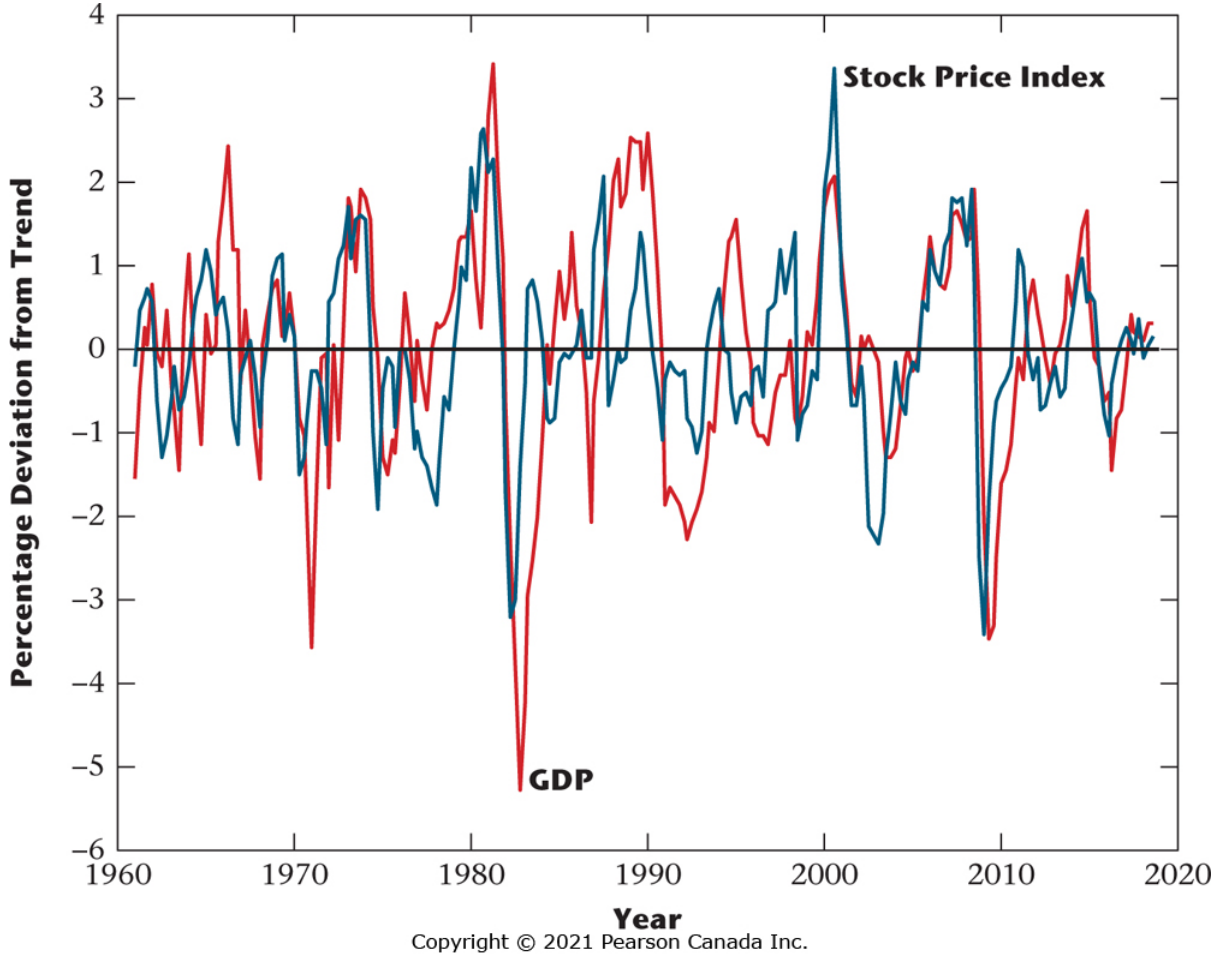
CYCLICALITY

- A variable is said to be **procyclical** if the deviation from its trend are positively correlated with the deviations of real GDP from its trend.
- A variable is said to be **countercyclical** if the deviation from its trend are negatively correlated with the deviations of real GDP from its trend.
- A variable is said to be **acyclical** if it is neither procyclical nor countercyclical
    - i.e., corrrelation coefficient is (close to) zero between fluctuations of the variable and real GDP.

LEADING AND LAGGING

- A variable is called a **leading variable** if it helps to predict the future path of real GDP.
- A variable is called a **lagging variable** if real GDP helps to predict the future path of the variable.
- Thus, to improve the forecasting of real GDP, it helps to find the leading variables.
- One leading variable is stock price index. Stock markets tend to quickly reflect information of market and the economy

Fluctuations of real GDP and TSE Composite Index



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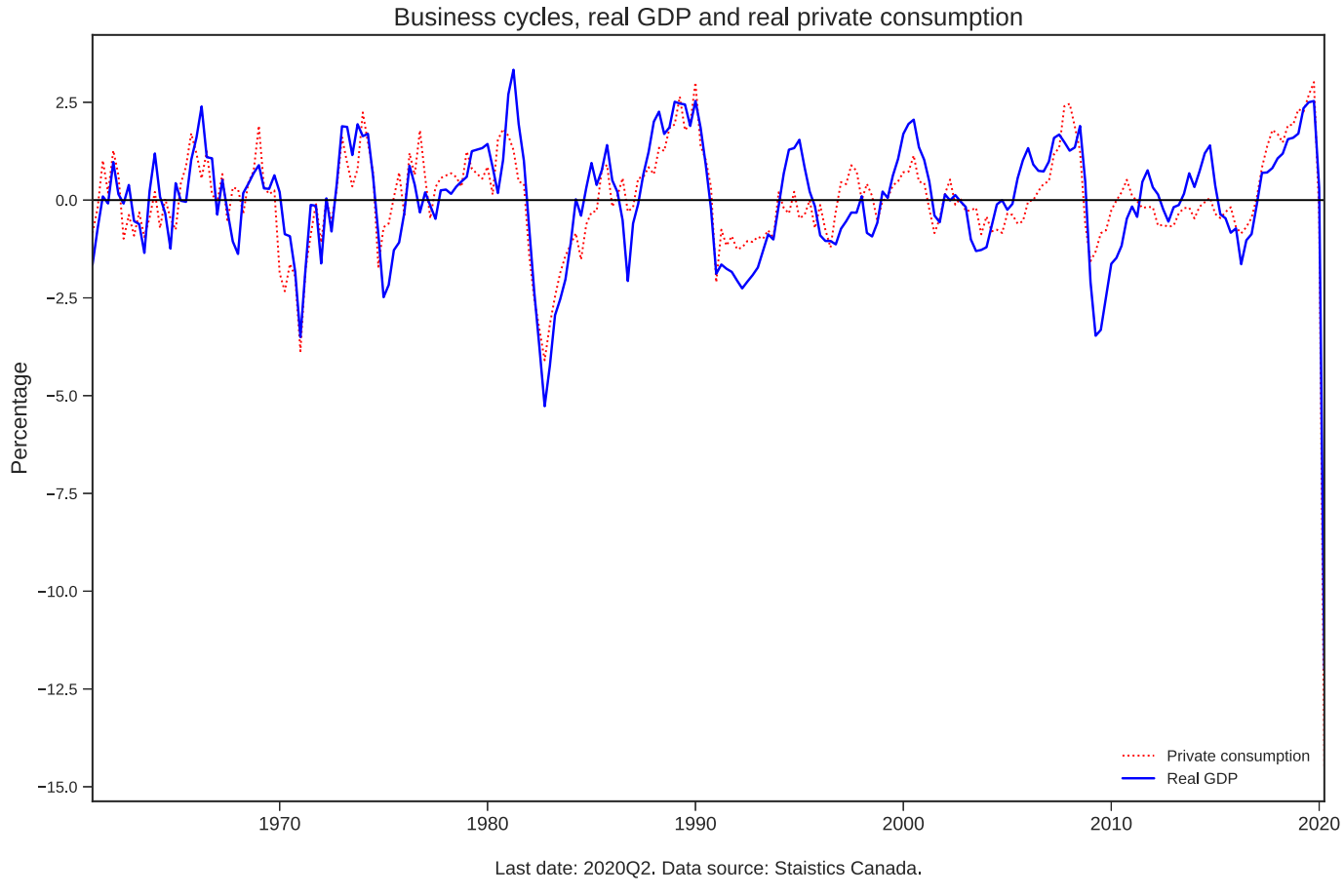
VOLATILITY

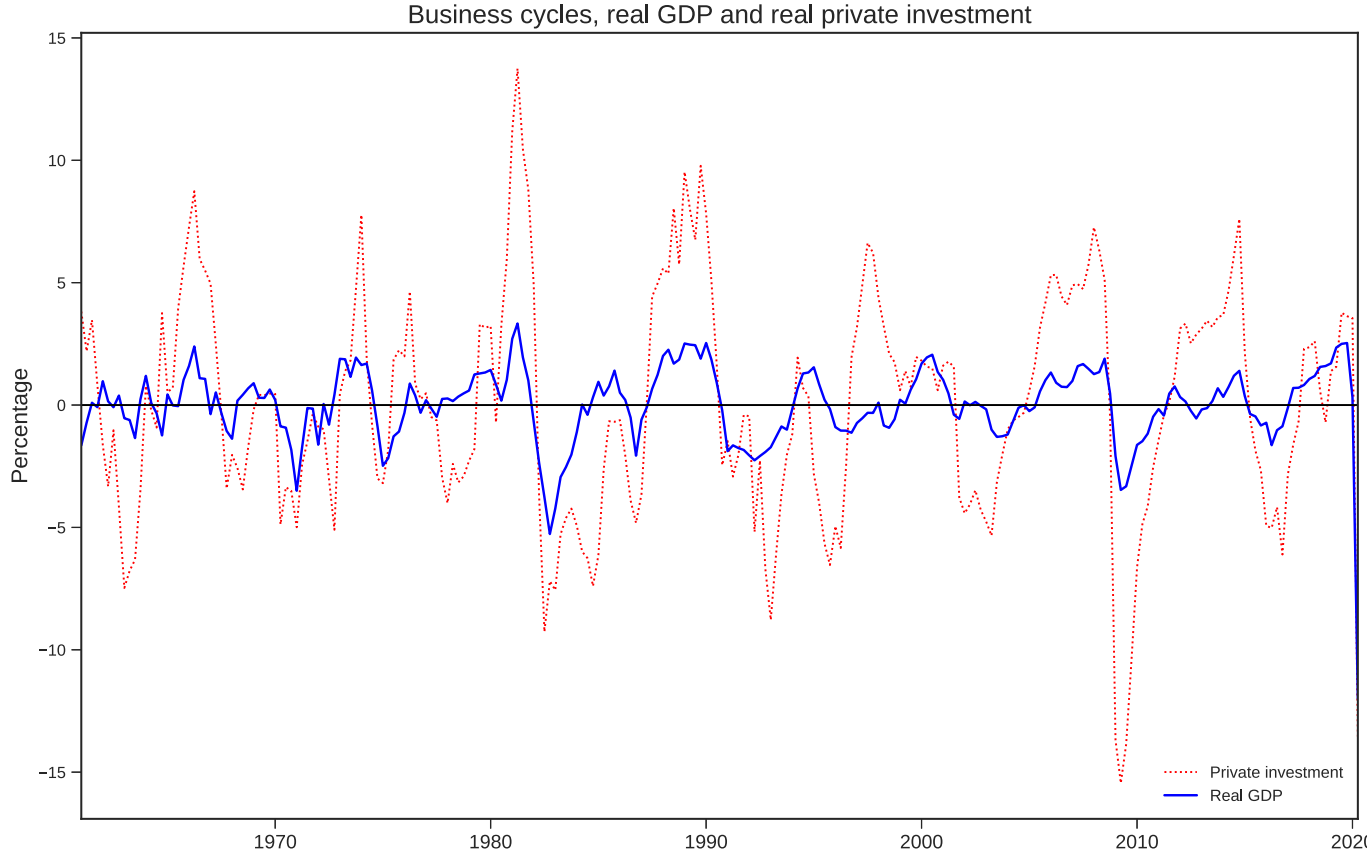
- **Volatility** measures the degree of variability of variables over the business cycle.
- We can measure volatility using the **standard deviation** of the deviations from trend.
    - Standar deviation is the square root of variance.
- Private investment is more volatile than GDP (see above figure).

2. SOME FACTS OF BUSINESS CYCLES IN CANADA

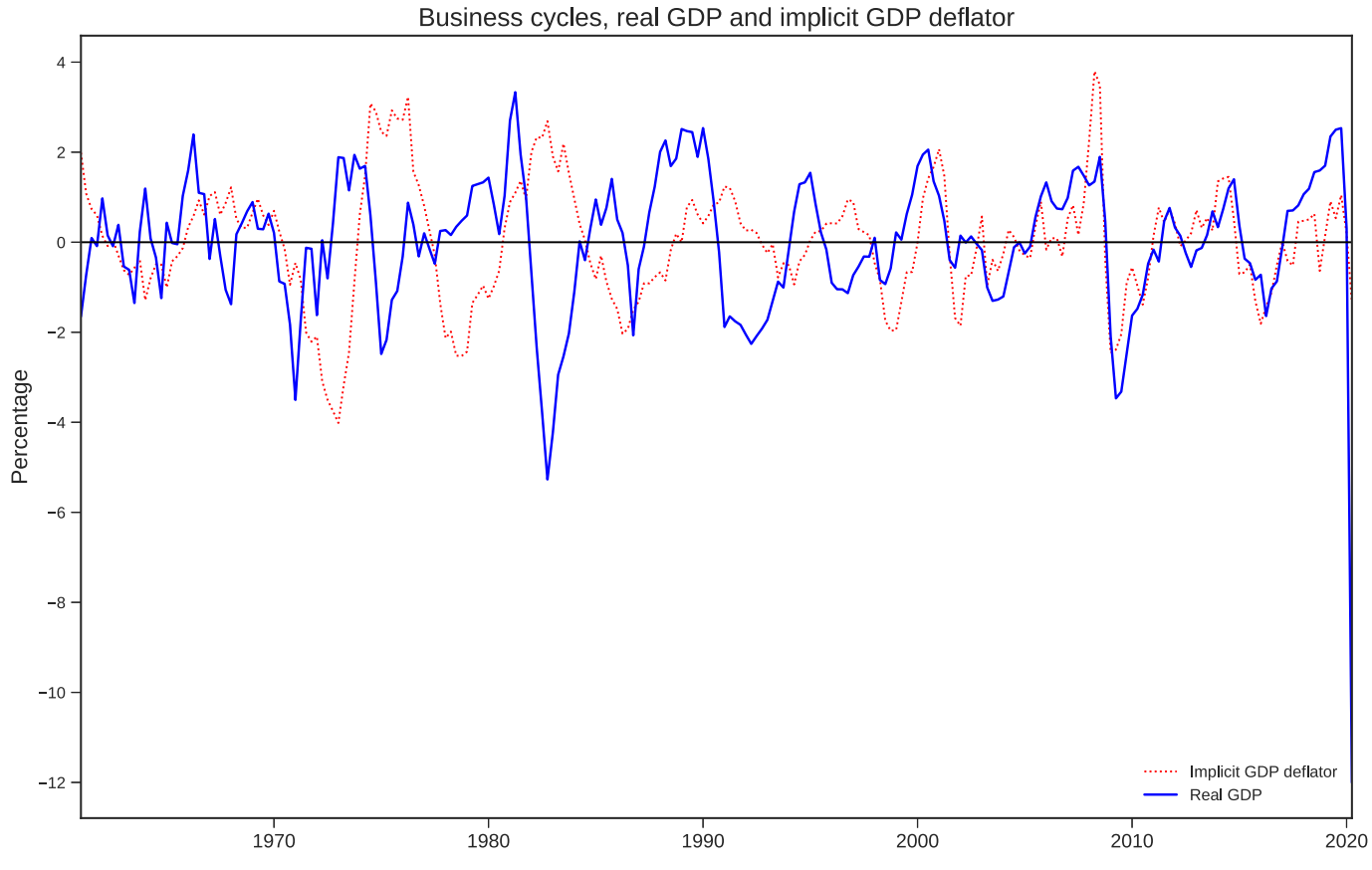
- The facts of interest are cyclicality, leading/lagging, and volatility of macroeconomic variables.
- Selected statistics of business cycles, 1962Q1 to 2020Q1.

Variable	First-order autocorrelation	Volatility	Contemporaneous correlation with output	Correlation with lag output	Correlation with lead output
GDP	0.84	1.33	1.00	0.84	0.84
Consumption	0.76	1.07	0.78	0.63	0.73
Investment	0.87	4.54	0.69	0.62	0.64
Price level	0.88	1.30	-0.03	0.04	-0.09
Inflation	0.79	1.49	0.35	0.34	0.28
Employment	0.93	1.15	0.71	0.75	0.57
ALP	0.66	0.77	0.33	0.19	0.39

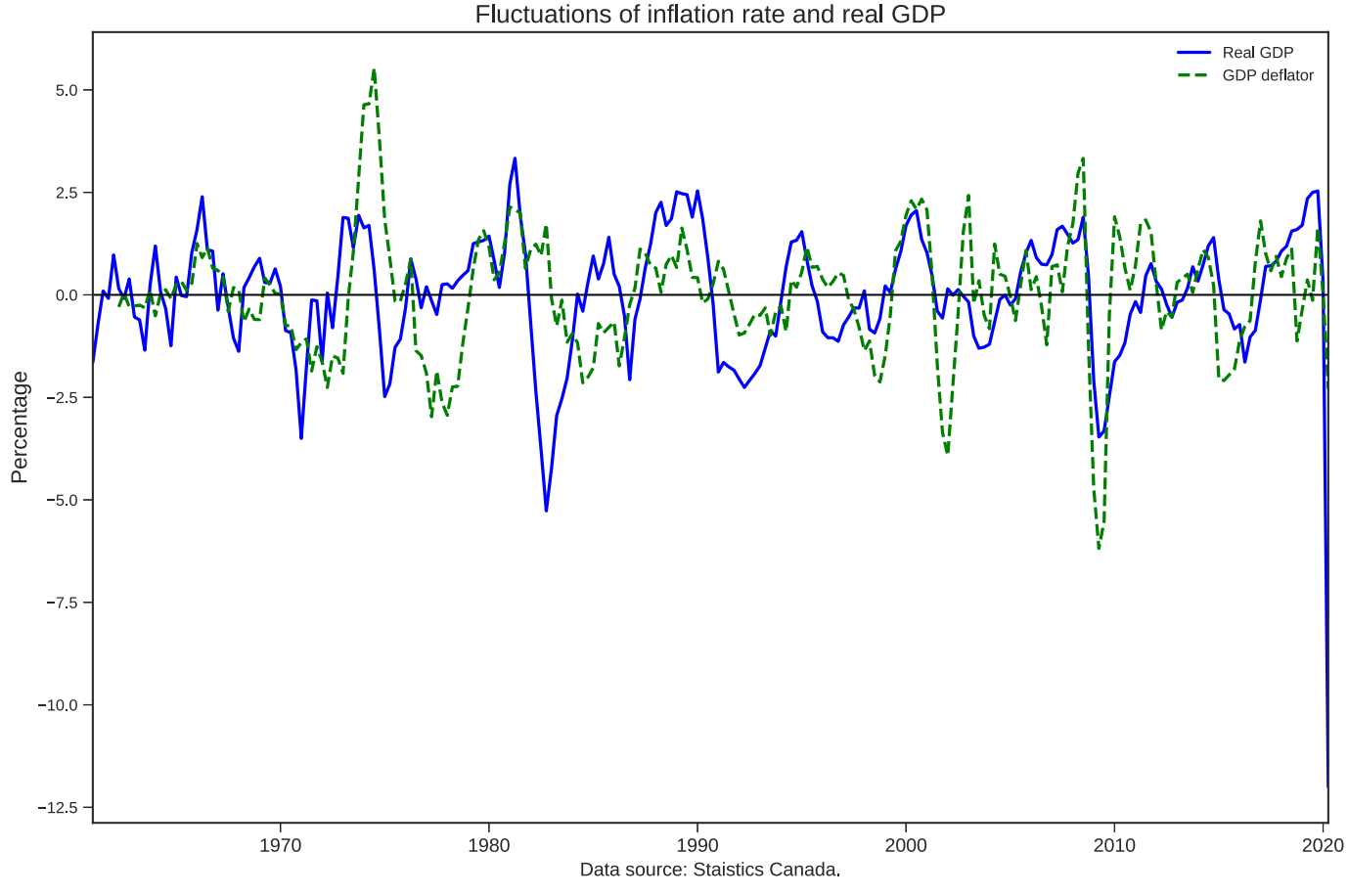

Business cycles, real GDP and real private consumption
Percentage
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
1970
1980
1990
2000
2010
2020
Private consumption
Real GDP
Last date: 2020Q2. Data source: Staistics Canada.

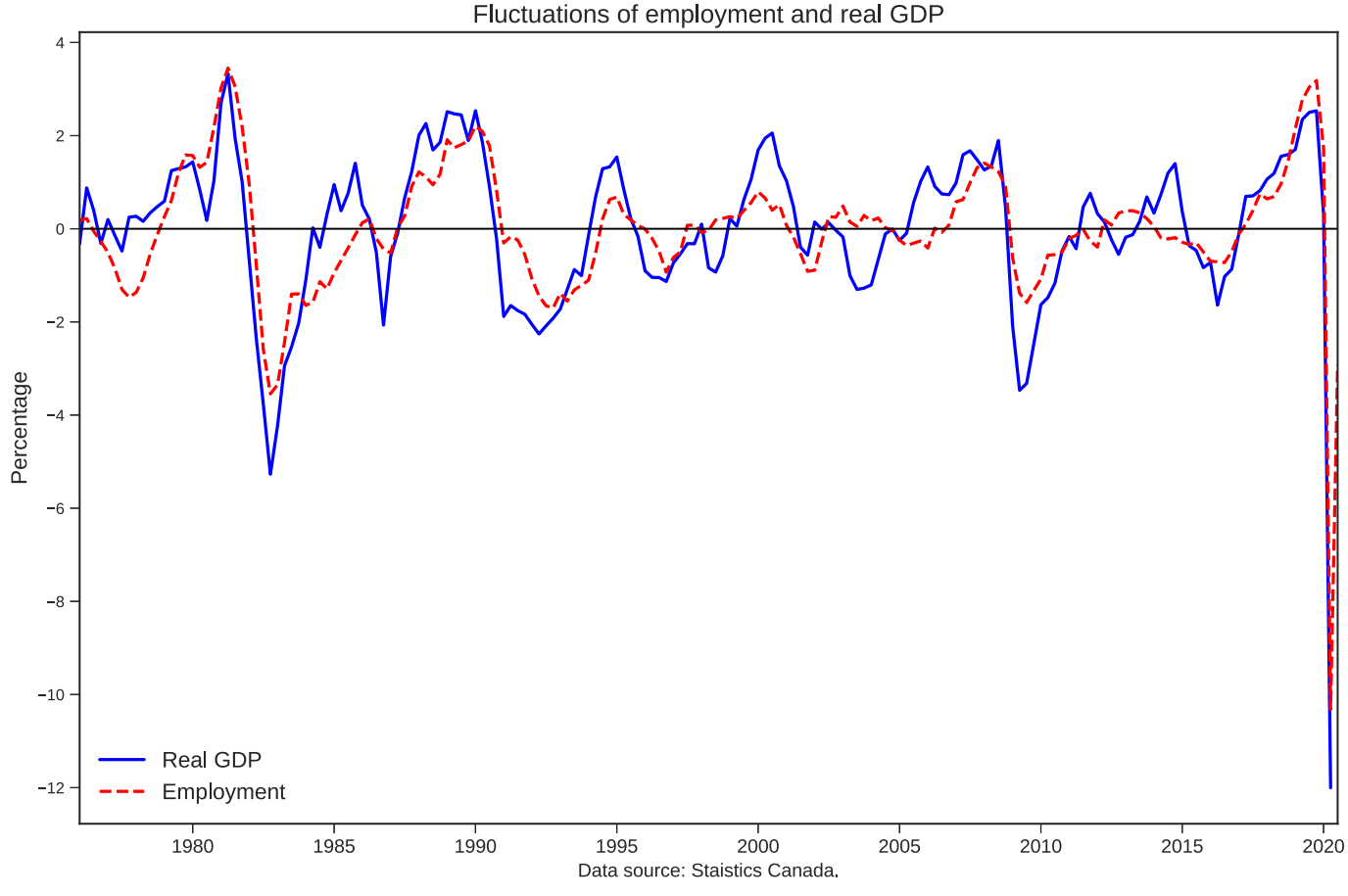


Last date: 2020Q2. Data source: Staistics Canada.

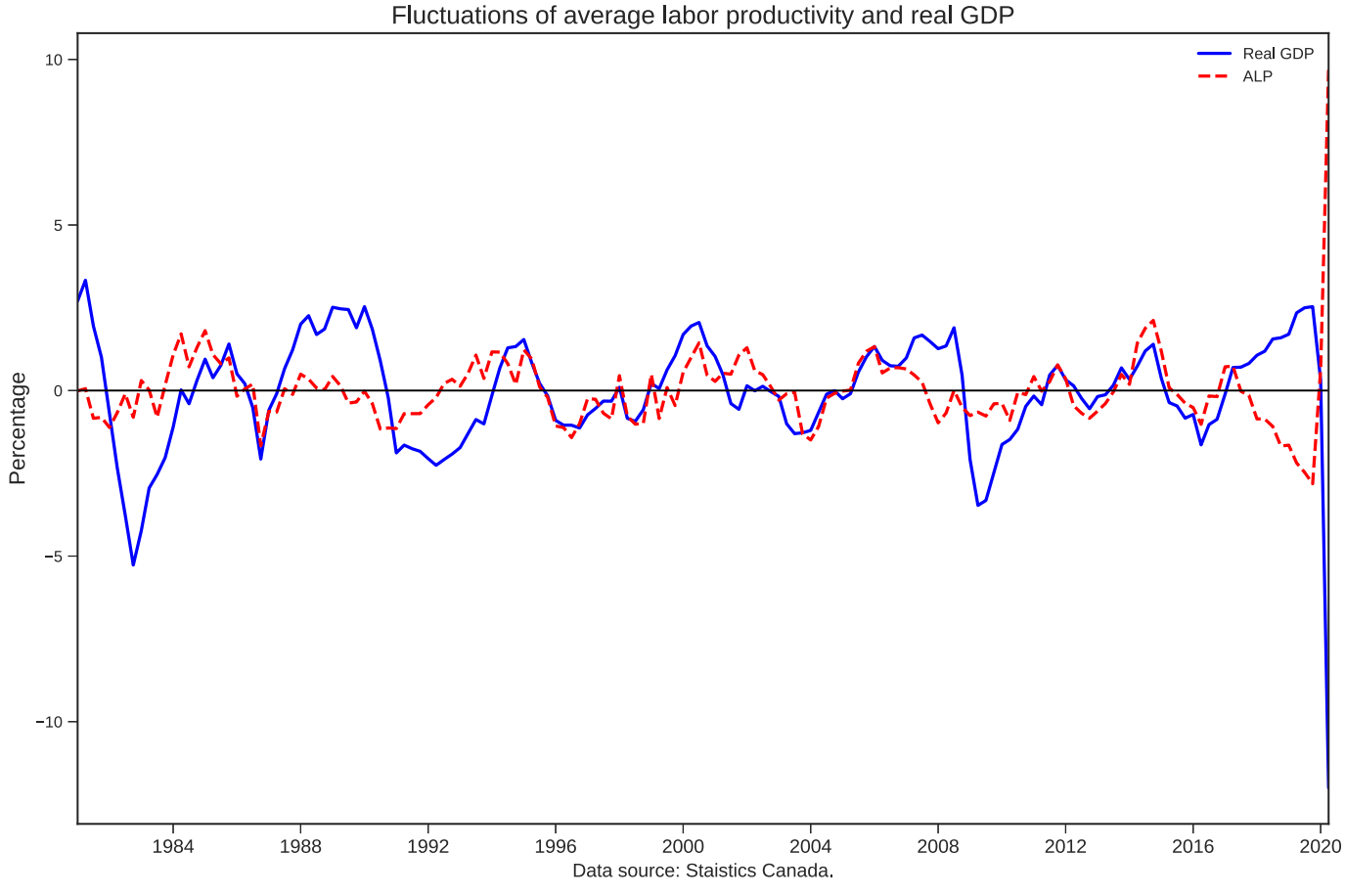


Last date: 2020Q2. Data source: Staistics Canada.


Fluctuations of inflation rate and real GDP
Real GDP
GDP deflator
Percentage
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
1970
1980
1990
2000
2010
2020
Data source: Staistics Canada.


Fluctuations of employment and real GDP
Percentage
4
2
0
−2
−4
−6
−8
−10
−12
1980
1985
1990
1995
2000
2005
2010
2015
2020
Real GDP
Employment
Data source: Staistics Canada.

- ALP, average labor productivity, is real GDP per hour worked, not real GDP per worker.



3. THE PHILLIPS CURVE

- The **Phillips curve** is an *observed* negative relationship between inflation rate and the unemployment rate.
- In A. W. Phillips' 1958 paper, inflation rate is the rate of changes in money wages.

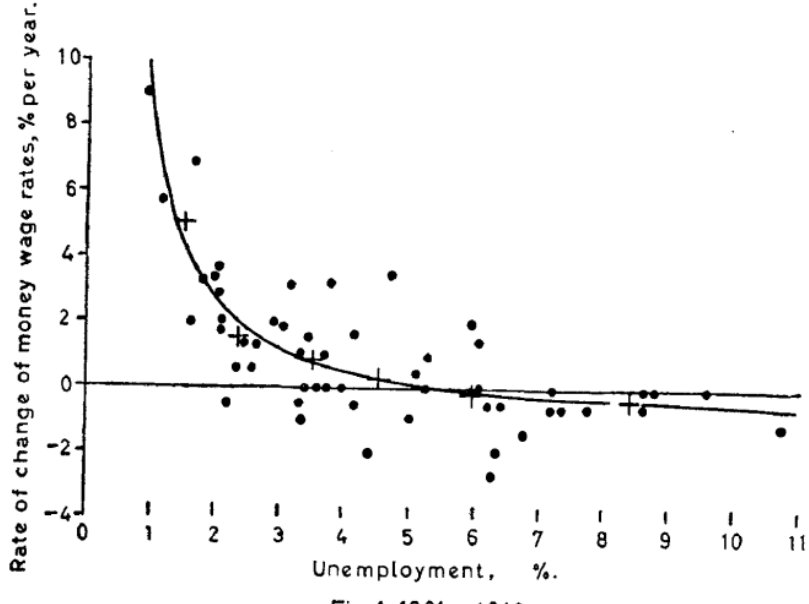
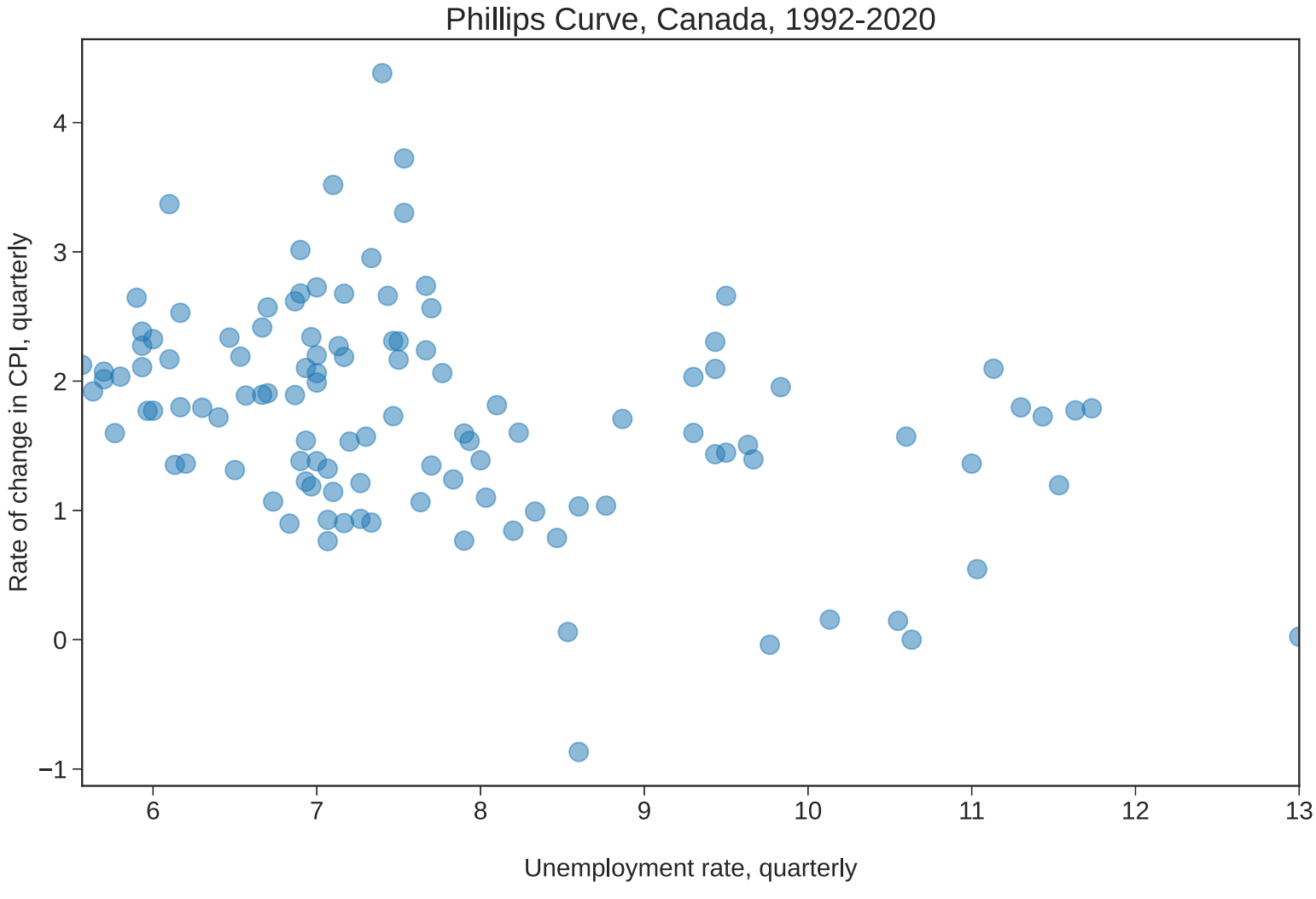
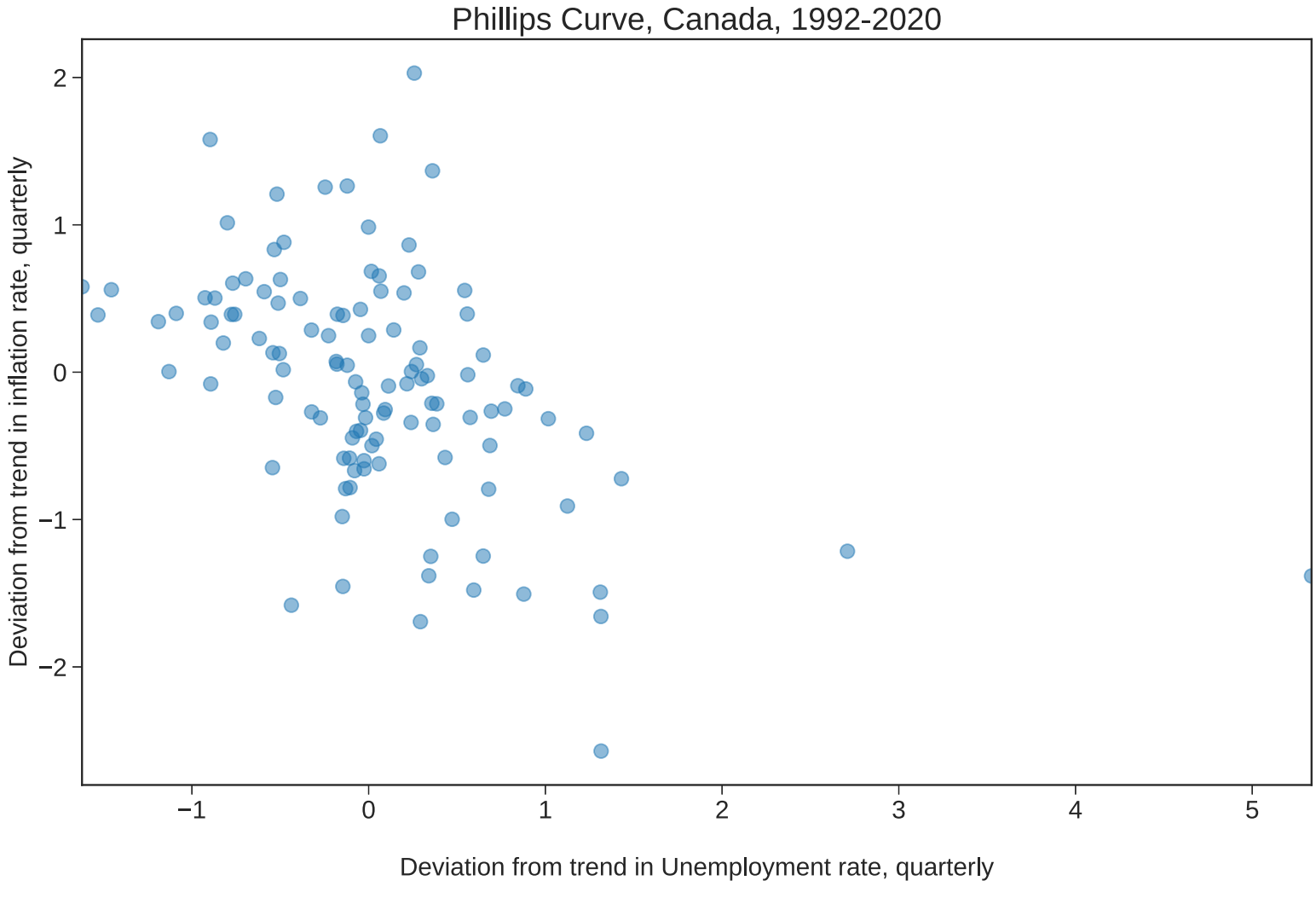


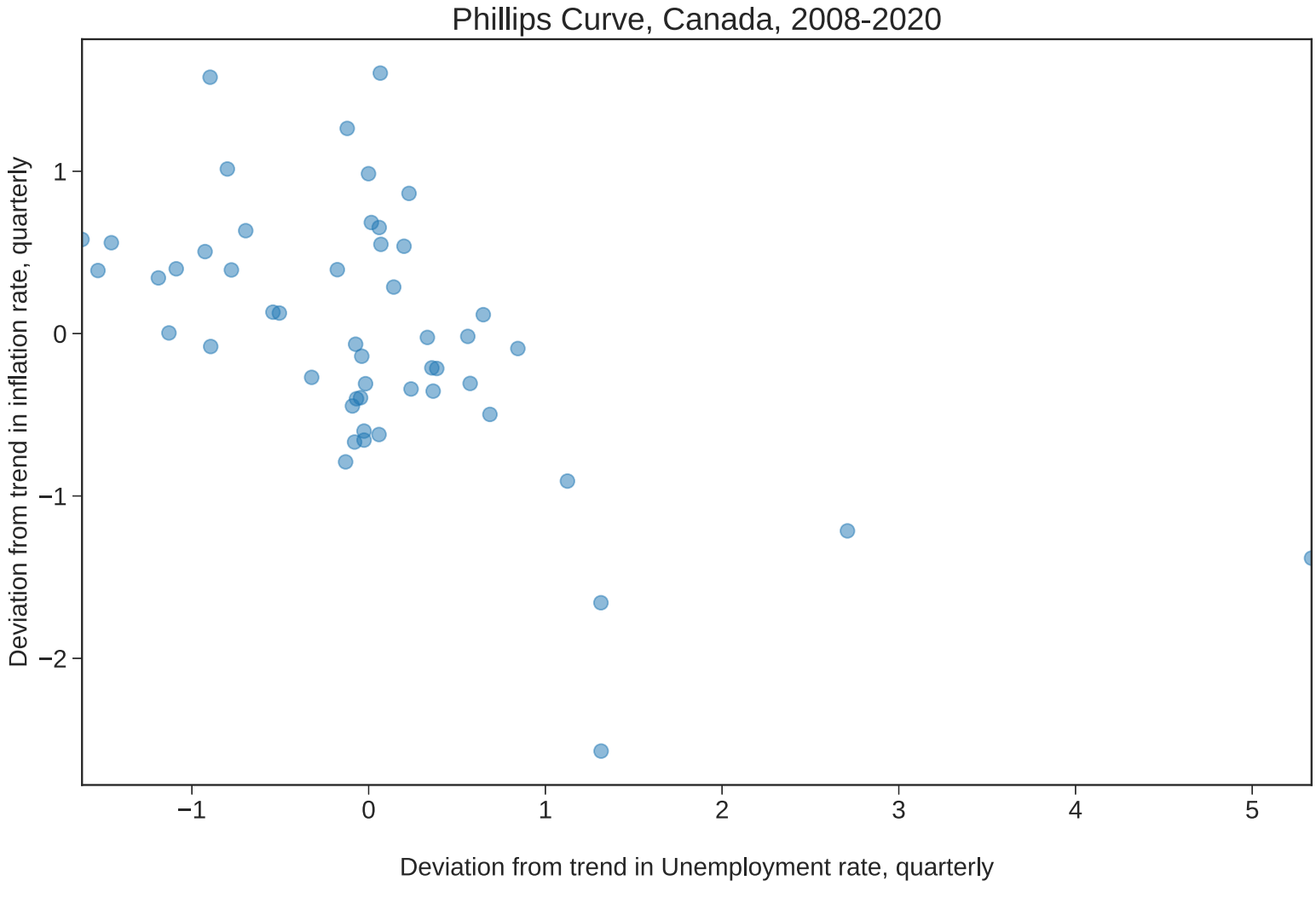
Fig.1. 1861 – 1913

Phillips (1958)

- Modern days, the Phillips curve refers to the relationship
    - between the inflation rate and the unemployment rate relative to the natural rate of unemployment.
    - between the deviation of inflation rate from trend and unemployment rate relative to the natural rate of unemployment.
- The textbook uses a broader definition, where the Phillips curve is the relationship between some inflation rate and the deviation from trend in *aggregate economic activity*.
- Data from recent experiences in Canada (and others for instance the U.S.) suggest
    - The observed Phillips curve is not a robust relationship between inflation and unemployment.
    - Some Phillips curve has disappeared post-2008.


Phillips Curve, Canada, 1992-2020
Rate of change in CPI, quarterly
Unemployment rate, quarterly
4
3
2
1
0
−1
6
7
8
9
10
11
12
13


Phillips Curve, Canada, 1992-2020
Deviation from trend in inflation rate, quarterly
Deviation from trend in Unemployment rate, quarterly
2
1
0
−1
−2
−1
0
1
2
3
4
5


Phillips Curve, Canada, 2008-2020
Deviation from trend in inflation rate, quarterly
Deviation from trend in Unemployment rate, quarterly
1
0
−1
−2
−1
0
1
2
3
4
5